

MTH203: Multivariate Calculus

Monsoon 2025

Name: _____

Roll No.: _____

Tutorial Quiz 11

Question:

1. Find the square roots of $3 + 4i$. (1)

2. Let

$$u(x, y) = \cos x \cosh y.$$

(a) Show that u is harmonic.

(b) Find a harmonic conjugate $v(x, y)$ and hence an analytic function

$$f(z) = u(x, y) + iv(x, y).$$

(2)

Please note: There is no fractional marking. Only integer marking.